

# INF221 - [Modern] Web Design & Development

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# Learning Outcome

By the end of this lecture, students should be able to:

- ❖ Outline principles, models, key elements and processes of information architecture for designing usable digital web products.

# Indicative Content

- ❖ Content strategy in Information Architecture (**IA**)
  - content planning and development
- ❖ Discuss key elements and models of IA
- ❖ Explore, describe tools that can be used for modelling IA in digital web products ie. Figma
- ❖ Describe and use effective ways of modelling the structure of the web content in web products.
  - Wireframes
  - Prototypes, etc.

# Background (1)

- ❖ **Structure** is key to easily **understanding** an artefact or any tangible object
  - House Building
  - Human Organ
  - Human Body - Overall
- ❖ Usually an artefact or an object is realized from some **design blueprint**
  - set of elements which guide implementers in the development or design of a tangible object
    - *consider a design blueprint from architectural/structural engineers provided to builders for building a house or structural engineers for building a highway road*
- ❖ In web design and development, web application's content must also be structured in way that users easily
  - understand and find what is provided on a website or web application.
- ❖ With clear and relevant content and understandable structure, users will be able to easily find web content.
  - enough reason for a user to visit our sight again and again

## Background (2)

“[The human] physical body can’t perform any task without the bones under [the] muscles and skin, which are designed to support those actions. In the same way, a site, app or system can’t deliver anything to [any user] unless its bone structure — it’s **Information Architecture** — is *specifically designed to support those tasks.*” **Udemy - Information Architecture (IA) Fundamentals**

## Background (3) - Definition

- ❖ IA refers to the structuring of a digital information space to provide users with simple, intuitive access to content and functions (Garrett, 2010, Rosenfeld and Morville, 2006)
  - significantly influences web apps usability and the user experience during navigation
  - helps to effectively communicate the purpose of the website or web application to users
    - without users' need for significant cognitive overload
  - **comprises:** organisation, navigation and labelling
- ❖ It is a visual representation of web application functional design elements, their hierarchy and relationship between them:
  - focuses on the end user's goals;
  - demonstrates a basic structure of the website or web application;
  - aimed at achieving less cognitive effort by its users.
- ❖ It is the practice of structuring, organising, and presenting content on digital platforms - websites, web applications, blogs, mobile applications etc.

# Goals

- User-Centered Organization: IA is designed with the user in mind.
- Findability: One of the primary objectives of IA is to ensure that information is easy to find.
- Scalability: As digital products grow and evolve, their IA needs to adapt seamlessly.
- Consistency Across Designs: IA promotes uniformity in how information is presented and organized across different sections of a product.
- Efficiency: An effective IA minimizes the cognitive load on users by making the layout and structure of the content clear and logical.

# Types of Data Organizing Structure

## ❖ The Hierarchical Tree Model

- It arranges content in a top-down manner, similar to a tree, where broad categories are at the top and subcategories branch out underneath.

## ❖ Sequential Structure:

- Organizes information in a linear or chronological order. This structure is ideal for guiding users through a step-by-step process

## ❖ Matrix Structure:

- The matrix structure allows information to be organized based on multiple criteria or dimensions, forming a grid-like arrangement.

## ❖ Network Structure:

- Connects different pieces of information like nodes in a web, where each node can link to multiple others.



# Types of Data Organizing Structure

- ❖ Database or Faceted Structure:
  - The faceted structure organizes information using multiple attributes or filters, allowing users to refine their search results.
- ❖ Alphabetical or A-Z Structure:
- ❖ Search-based Structure:
- ❖ Spatial Structure:
  - Spatial structures organize information in a way that mirrors physical environments or virtual spaces.

# User Centred IA: Card Sorting

- ❖ Card sorting is a user research approach in Information Architecture (IA) that is used to investigate how people categorize and organize information in a system.
- ❖ Card sorting entails people categorizing things or themes written on physical cards or in digital media into meaningful groups. These groupings can then influence the information structure in the final product.
- ❖ Steps include: preparation, recruitment, making cards, instructions, data collection and analysis

# References

- ❖ <https://www.udemy.com/course/ia-fundamentals/?couponCode=LETSLEARNNOW>, accessed on 9th April 2024.
- ❖ J.J. Garrett (2010) The elements of user experience.
- ❖ [Information Architecture: A Complete Guide For Beginners | GeeksforGeeks](#)